

ISABELLA CHANDLER

BIOMEDICAL ENGINEERING | MEDICAL DEVICES | QUALITY ENGINEERING | HEALTHCARE TECHNOLOGY

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Biomedical Engineering student with hands-on experience in quality-focused device preparation, technical troubleshooting, biomedical instrumentation, embedded systems, and prototyping. Quality Science Education Certificate holder with training in FDA quality regulations, GMP, validation, risk management, and data integrity, with current study in medical imaging, tissue engineering, biomechanics, and nanotechnology.

EDUCATION & CERTIFICATION

Lawrence Technological University | B.S. Biomedical Engineering | Southfield, MI

Expected May 2027

GPA: 3.16/4.00 | **Quality Science Education Certificate**, Pathway for Patient Health - Completed Aug 2025

Quality training: FDA quality regulations, GMP, risk assessment, validation, data integrity, and change control.

Current coursework (Fall 2026): Foundations of Medical Imaging, Tissue Engineering + Lab, Biomechanics + Lab, Introduction to Nanotechnology, BME Projects I.

Additional relevant coursework: Bioinstrumentation, Biotransport, Microfluidics, Wearable Technology Design Studio, BME Ethics & Regulations.

Honors: LTU Honor Roll - Fall 2024 and Summer 2025.

TECHNICAL SKILLS

Engineering & Analysis: MATLAB, COMSOL, AutoCAD/CAD, 3D printing, prototyping, soldering, sensor calibration and testing

Programming & Embedded: Python/CircuitPython, C#, BLE, microcontrollers, servo control

Quality & Regulatory: quality control and inspection, validation fundamentals, risk assessment, GMP/FDA quality principles, technical documentation, data integrity

Tools & Communication: Microsoft Excel, Microsoft 365, Spanish (Advanced)

EXPERIENCE & LEADERSHIP

Help Desk Assistant | Lawrence Technological University | Southfield, MI

Jan 2025 - Present

- Prepared 2,000+ university laptops for student distribution through hardware inspection, cleaning, functional checks, and deployment-readiness verification; performed OS imaging across Lenovo T1/T2/T3, x360, and Fujitsu systems.
- Independently maintained summer device-staging operations, organizing labeled inventory and location/status controls, charger and accessory preparation, and charging capacity to support high-volume semester turnaround.
- Provided front-line technology support to students, faculty, and staff while maintaining student-facing printer readiness through troubleshooting, routine checks, and print runs.

Director of Programming, Executive Board | LTU Student Government | Southfield, MI

Aug 2025 - May 2026

- Directed campus programming and coordinated planning for a \$10,000+ end-of-year celebration, including vendor quotes, rentals, facilities, schedules, and the university event calendar with Student Life and Senate.
- Led event execution across food trucks, inflatables, games, setup, safety/logistics, and day-of operations; previously served as Student Body Senator and Safety, Health & Mental Health Committee Leader.

Biomedical Engineering Lab Assistant | Lawrence Technological University | Southfield, MI

Apr - Aug 2024

- Prepared BME teaching and lab spaces for summer-camp activities; maintained Excel-based inventory, organized supplies and equipment, and supported recruitment programming for prospective students.

SELECTED ENGINEERING PROJECTS

Mopec Tissue Insight | *Company-Sponsored Senior Design* | 2026-2027

- Contribute to early-stage development of an AI-assisted gross-tissue imaging workflow by reviewing datasets and defining image-QC, preprocessing, annotation, train/validation/test, and expert-review requirements.

HeadUp Posture-Feedback Wearable | *Bioinstrumentation*

- Designed and tested an accelerometer-based posture wearable using user calibration, threshold logic, and haptic/visual feedback; evaluated detection performance, activation delay, comfort, and feedback strength.

ConnectUs Accessible Dice Tower | *Entrepreneurial Engineering Design*

- Translated customer needs into engineering requirements and iterated an accessible prototype using Arduino, servo control, 3D printing, soldering/wiring, material selection, testing, and user feedback.